



IMO Shortlist 2012

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by teps, lyukhson, delegat, SpectralS, mathmdmb, Tonci

– Algebra

- A1** Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that, for all integers a, b, c that satisfy $a + b + c = 0$, the following equality holds:

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a).$$

(Here $\mathbb{Z}$ denotes the set of integers.)

Proposed by Liam Baker, South Africa

- A2** Let $\mathbb{Z}$ and $\mathbb{Q}$ be the sets of integers and rationals respectively.
a) Does there exist a partition of $\mathbb{Z}$ into three non-empty subsets A, B, C such that the sets $A + B, B + C, C + A$ are disjoint?
b) Does there exist a partition of $\mathbb{Q}$ into three non-empty subsets A, B, C such that the sets $A + B, B + C, C + A$ are disjoint?

Here $X + Y$ denotes the set $\{x + y : x \in X, y \in Y\}$, for $X, Y \subseteq \mathbb{Z}$ and for $X, Y \subseteq \mathbb{Q}$.

- A3** Let $n \geq 3$ be an integer, and let $a_2, a_3, \dots, a_n$ be positive real numbers such that $a_2 a_3 \cdots a_n = 1$. Prove that

$$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

Proposed by Angelo Di Pasquale, Australia

- A4** Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

- A5** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy the conditions

$$f(1 + xy) - f(x + y) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R},$$

and $f(-1) \neq 0$.

- A6** Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function, and let f^m be f applied m times. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k . Prove that the sequence $k_1, k_2, \dots$ is unbounded.

Proposed by Palmer Mebane, United States

- A7** We say that a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is a metapolynomial if, for some positive integers m and n , it can be represented in the form

$$f(x_1, \dots, x_k) = \max_{i=1, \dots, m} \min_{j=1, \dots, n} P_{i,j}(x_1, \dots, x_k),$$

where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

– Combinatorics

- C1** Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y , and replaces the pair (x, y) by either $(y + 1, x)$ or $(x - 1, x)$. Prove that she can perform only finitely many such iterations.

Proposed by Warut Suksompong, Thailand

- C2** Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \dots, n\}$ such that the sums of the different pairs are different integers not exceeding n ?

- C3** In a 999×999 square table some cells are white and the remaining ones are red. Let T be the number of triples (C_1, C_2, C_3) of cells, the first two in the same row and the last two in the same column, with C_1, C_3 white and C_2 red. Find the maximum value T can attain.

Proposed by Merlijn Staps, The Netherlands

- C4** Players A and B play a game with $N \geq 2012$ coins and 2012 boxes arranged around a circle. Initially A distributes the coins among the boxes so that there is at least 1 coin in each box. Then the two of them make moves in the order $B, A, B, A, \dots$ by the following rules:

(a) On every move of his B passes 1 coin from every box to an adjacent box.

(b) On every move of hers A chooses several coins that were *not* involved in B 's previous move and are in different boxes. She passes every coin to an adjacent box.

Player A 's goal is to ensure at least 1 coin in each box after every move of hers, regardless of how B plays and how many moves are made. Find the least N that enables her to succeed.

- C5** The columns and the row of a $3n \times 3n$ square board are numbered $1, 2, \dots, 3n$. Every square (x, y) with $1 \leq x, y \leq 3n$ is colored asparagus, byzantium or citrine according as the modulo 3 remainder of $x + y$ is 0, 1 or 2 respectively. One token colored asparagus, byzantium or citrine is placed on each square, so that there are $3n^2$ tokens of each color.

Suppose that one can permute the tokens so that each token is moved to a distance of at most d from its original position, each asparagus token replaces a byzantium token, each byzantium token replaces a citrine token, and each citrine token replaces an asparagus token. Prove that it is possible to permute the tokens so that each token is moved to a distance of at most $d + 2$ from its original position, and each square contains a token with the same color as the square.

- C6** The *liar's guessing game* is a game played between two players A and B . The rules of the game depend on two positive integers k and n which are known to both players.

At the start of the game A chooses integers x and N with $1 \leq x \leq N$. Player A keeps x secret, and truthfully tells N to player B . Player B now tries to obtain information about x by asking player A questions as follows: each question consists of B specifying an arbitrary set S of positive integers (possibly one specified in some previous question), and asking A whether x belongs to S . Player B may ask as many questions as he wishes. After each question, player A must immediately answer it with *yes* or *no*, but is allowed to lie as many times as she wants; the only restriction is that, among any $k + 1$ consecutive answers, at least one answer must be truthful.

After B has asked as many questions as he wants, he must specify a set X of at most n positive integers. If x belongs to X , then B wins; otherwise, he loses. Prove that:

1. If $n \geq 2^k$, then B can guarantee a win.
2. For all sufficiently large k , there exists an integer $n \geq (1.99)^k$ such that B cannot guarantee a win.

Proposed by David Arthur, Canada

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- C7** There are given 2^{500} points on a circle labeled $1, 2, \dots, 2^{500}$ in some order. Prove that one can choose 100 pairwise disjoint chords joining some of these points so that the 100 sums of the pairs of numbers at the endpoints of the chosen chord are equal.
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– Geometry

- G1** Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC . Prove that M is the midpoint of ST .

(The excircle of ABC opposite the vertex A is the circle that is tangent to the line segment BC , to the ray AB beyond B , and to the ray AC beyond C .)

Proposed by Evangelos Psychas, Greece

- G2** Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.
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- G3** In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the

circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

- G4** Let ABC be a triangle with $AB \neq AC$ and circumcenter O . The bisector of $\angle BAC$ intersects BC at D . Let E be the reflection of D with respect to the midpoint of BC . The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.

- G5** Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK .

Show that $MK = ML$.

Proposed by Josef Tkadlec, Czech Republic

- G6** Let ABC be a triangle with circumcenter O and incenter I . The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

- G7** Let $ABCD$ be a convex quadrilateral with non-parallel sides BC and AD . Assume that there is a point E on the side BC such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point F on the side AD such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if AB is parallel to CD .

- G8** Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω . Denote by P the foot of the perpendicular from the center of ω to ℓ . The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P . Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P .

Proposed by Cosmin Pohoata, Romania

– Number Theory

- N1** Call admissible a set A of integers that has the following property:
If $x, y \in A$ (possibly $x = y$) then $x^2 + kxy + y^2 \in A$ for every integer k .
Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

Proposed by Warut Suksompong, Thailand

- N2** Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

N3 Determine all integers $m \geq 2$ such that every n with $\frac{m}{3} \leq n \leq \frac{m}{2}$ divides the binomial coefficient $\binom{n}{m-2n}$.

N4 An integer a is called friendly if the equation $(m^2 + n)(n^2 + m) = a(m - n)^3$ has a solution over the positive integers.
a) Prove that there are at least 500 friendly integers in the set $\{1, 2, \dots, 2012\}$.
b) Decide whether $a = 2$ is friendly.

N5 For a nonnegative integer n define $\text{rad}(n) = 1$ if $n = 0$ or $n = 1$, and $\text{rad}(n) = p_1 p_2 \cdots p_k$ where $p_1 < p_2 < \cdots < p_k$ are all prime factors of n . Find all polynomials $f(x)$ with nonnegative integer coefficients such that $\text{rad}(f(n))$ divides $\text{rad}(f(n^{\text{rad}(n)}))$ for every nonnegative integer n .

N6 Let x and y be positive integers. If $x^{2^n} - 1$ is divisible by $2^n y + 1$ for every positive integer n , prove that $x = 1$.

N7 Find all positive integers n for which there exist non-negative integers $a_1, a_2, \dots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

Proposed by Dusan Djukic, Serbia

N8 Prove that for every prime $p > 100$ and every integer r , there exist two integers a and b such that p divides $a^2 + b^5 - r$.
